COMP2300/6300

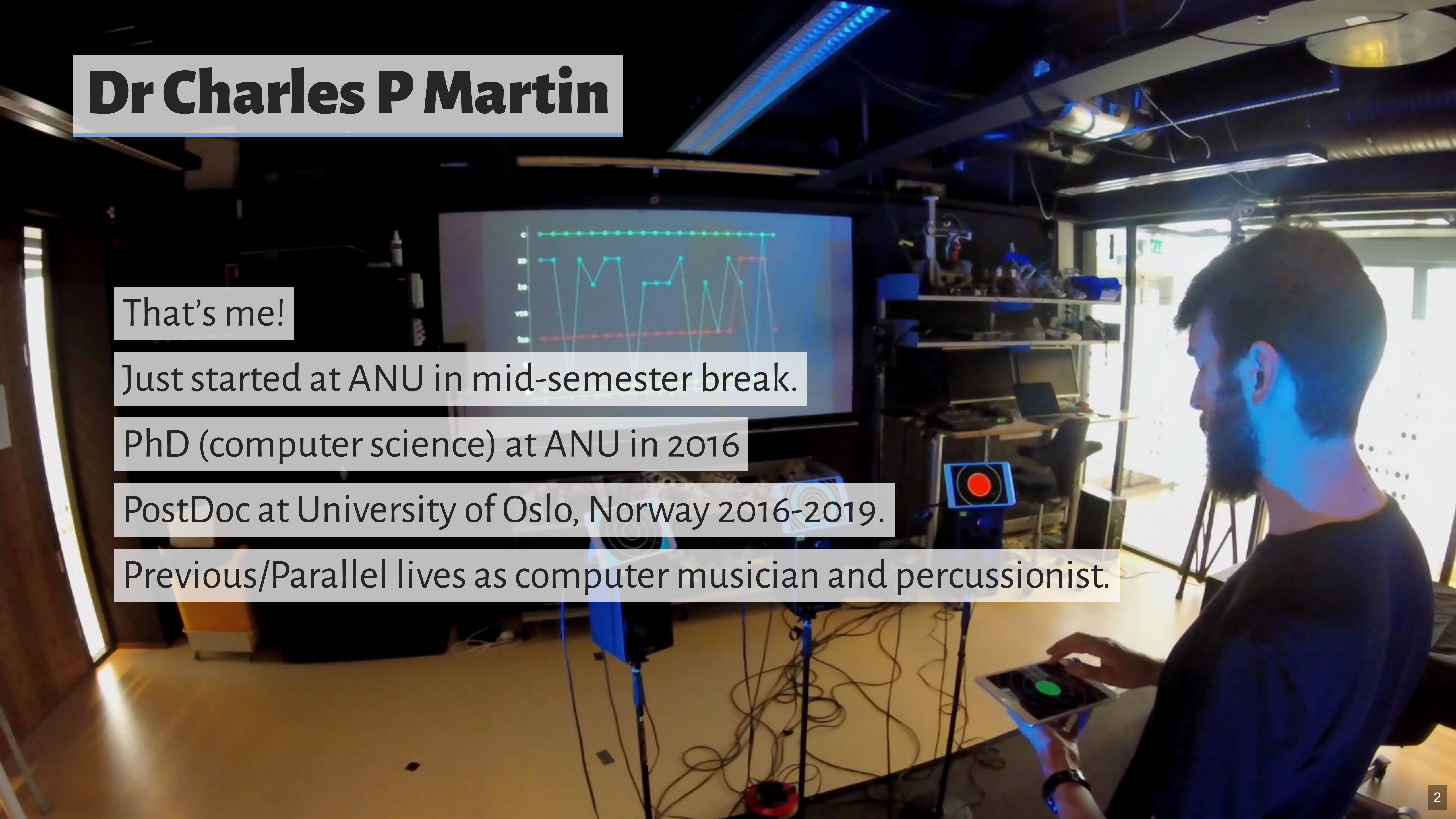
Computer Organisation & Program Execution

Dr Uwe Zimmer

Dr Charles Martin

Semester 1, 2019

Dr Charles P Martin



That's me!

Just started at ANU in mid-semester break.

PhD (computer science) at ANU in 2016

PostDoc at University of Oslo, Norway 2016-2019.

Previous/Parallel lives as computer musician and percussionist.

Week 7: Interrupts; Digital Synthesis

info

- Assignment 1 marks are out on streams.
- Mid-semester exam marks are out. (see stats)
- Assignment 2 is out, please get started! **Push Early, Push Often!**
- No lecture this Thursday (25 April, ANZAC day public holiday)
- No tutorials this Thursday, go to another session this week.

talk

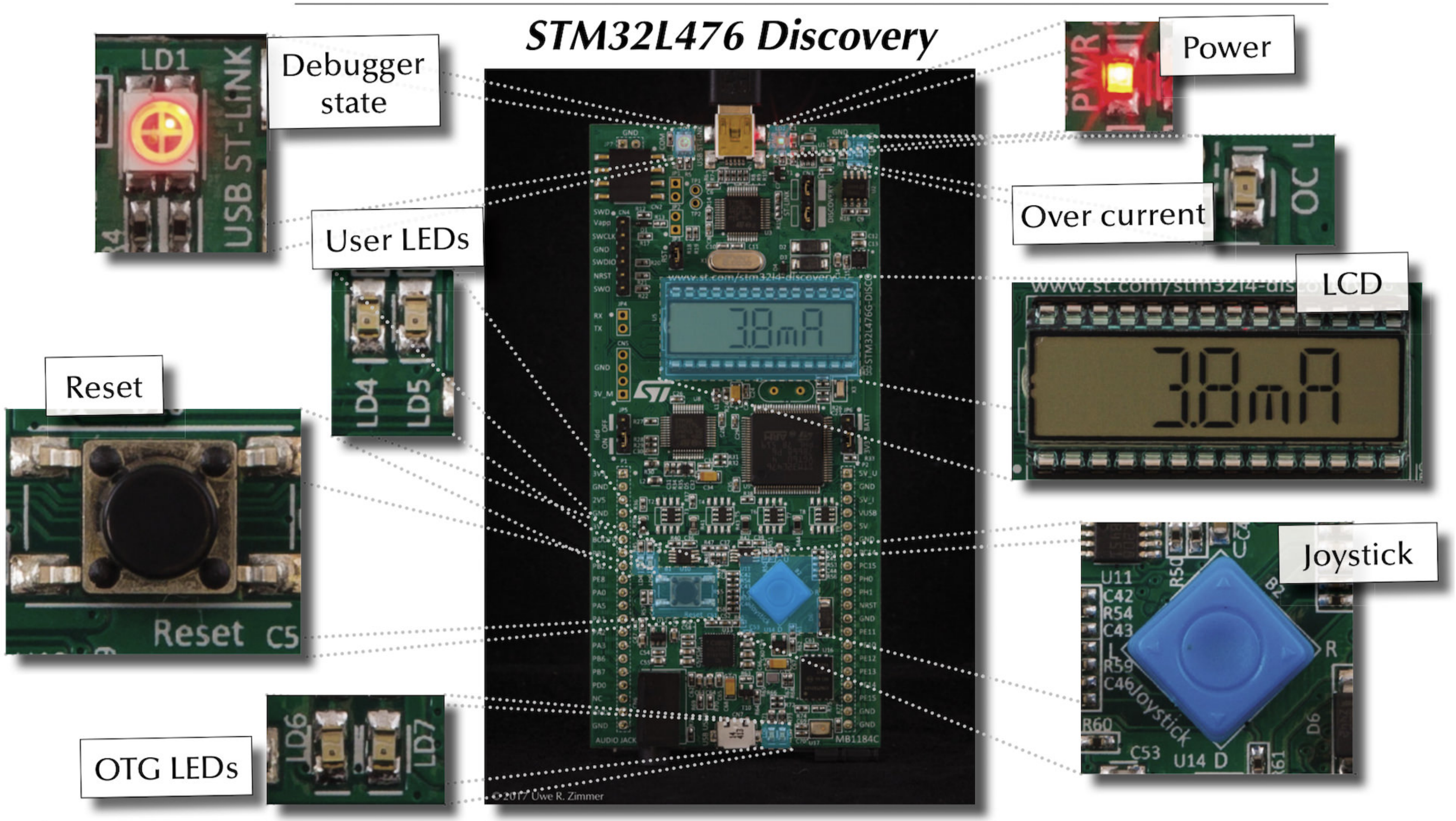
What's an interrupt? Explain like I'm 5.

- What could interrupts be used for?
- Can you prevent an interrupt?

You’ve probably experienced an interrupt (exception) already!

What about “Usage Fault”?

Connecting to the world!



How do we configure an interrupt?

1. Need to enable the interrupt.
2. Need to define the handler function.
3. Need to configure hardware (if using MCU features).

Where are interrupt vectors defined?

Have a look in `startup_stm32l476xx.S`:

```
g_pfnVectors:
  .word _estack
  .word Reset_Handler
  .word NMI_Handler
  .word HardFault_Handler
  .word MemManage_Handler
  ...
```

All the interrupt vectors are named here, and linked to a default handler.

How do we take over an interrupt handler?

Need to redefine one of the handler functions. E.g. for `EXTI0_IRQHandler`:

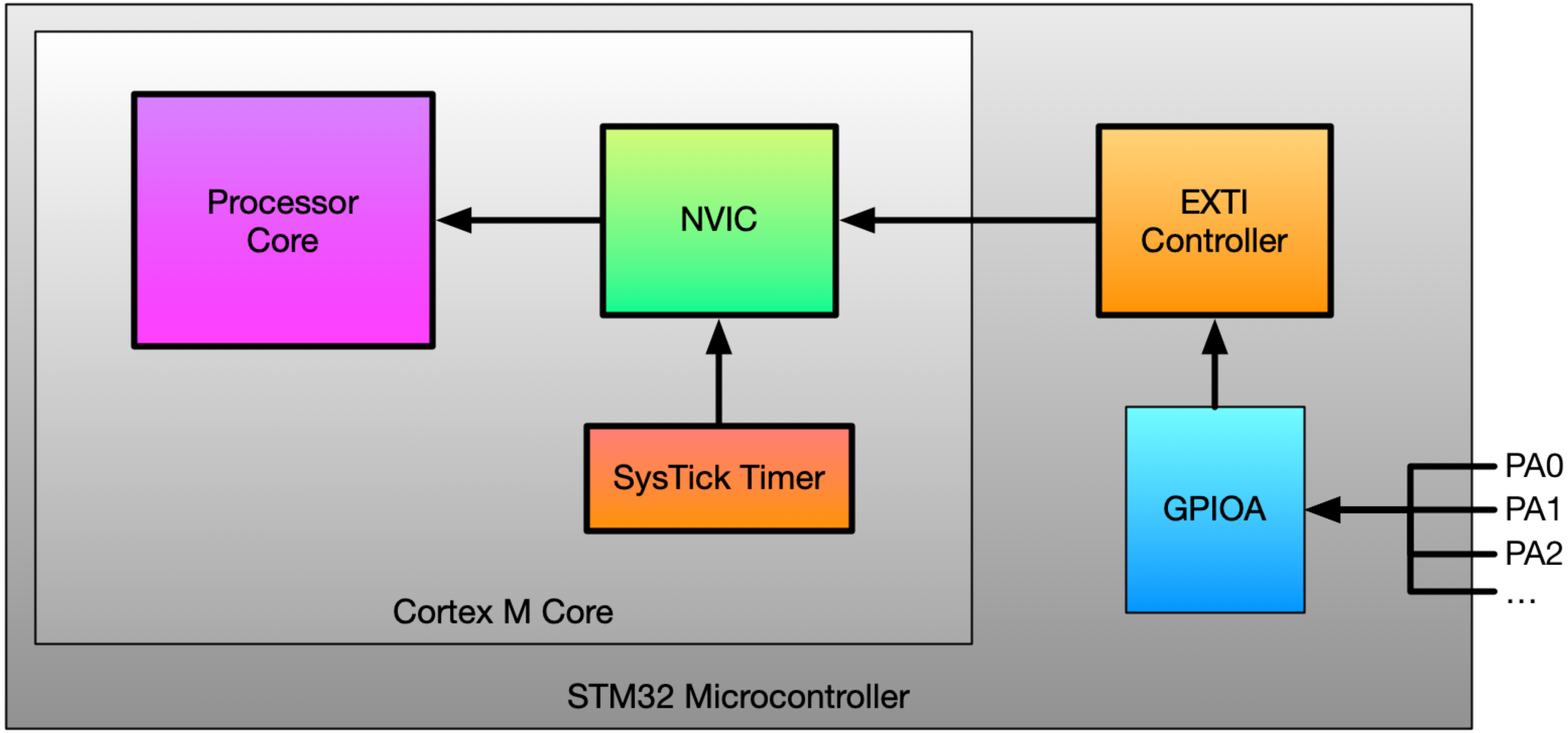
```
.global EXTI0_IRQHandler
.type EXTI0_IRQHandler, %function
EXTI0_IRQHandler:
  @ do something!
  bx lr
.size EXTI0_IRQHandler, .-EXTI0_IRQHandler
```

So I want to use the joystick...

Isn't that similar to activating the LEDs?

A map to the joystick...

The joystick centre button is connected to pin `PA0` on your disco board.

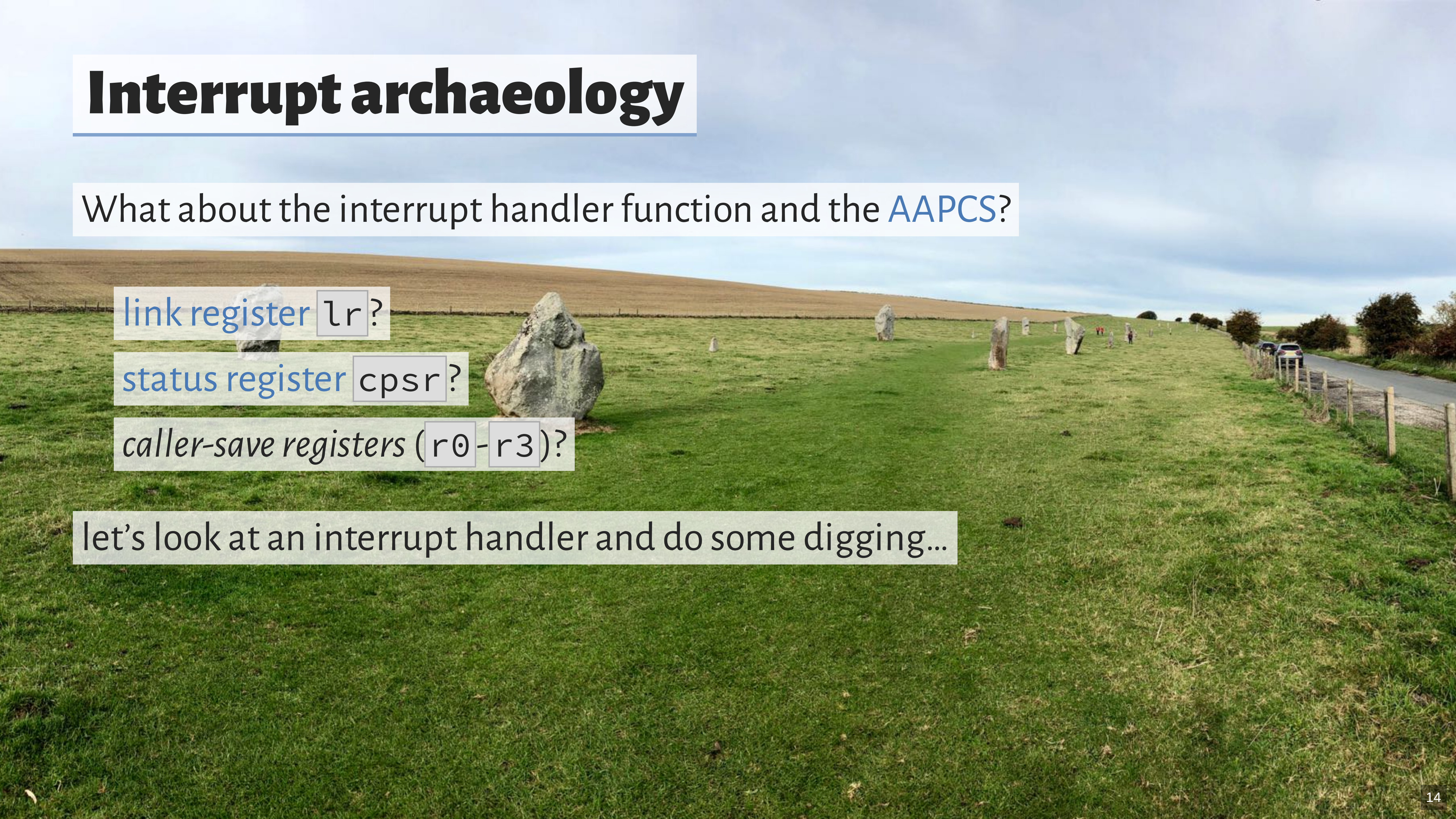


Connecting it all up

We need to configure `GPIOA0` as an input, configure the external interrupt controller (EXTI), *and* configure the NVIC to make this happen. This is a bit fussy!

- Enable `GPIOA` and `SYSCFG` clocks.
- Set `PA0` to input, and activate pull-down resistor
- Set `PA0` as source for `EXTI0` interrupt.
- Enable `EXTI0` interrupts and set `EXTI0` to rising edge trigger.
- Enable `EXTI0` interrupts in the `NVIC`.

Let's do it.



Interrupt archaeology

What about the interrupt handler function and the AAPCS?

- link register `lr`?
- status register `cpsr`?
- *caller-save registers* (`r0`-`r3`)?

let's look at an interrupt handler and do some digging...

talk

What's the relationship between the concepts of **interrupts** and **concurrency**?

Using Synchronisation Primitives

What's the deal with **ldrex** and **strex**?

LDREX

```
ldr r0, =label_in_data_section
ldrex r1, [r0]
```

`ldrex` loads `r1` with the memory that `r0` is pointing to, and sets that memory address in the "local exclusive monitor".

It doesn't do any checking of the exclusive monitor before doing so!

Multi-processor systems also have a "global exclusive monitor"... not covered here!

STREX

```
ldr r0, =label_in_data_section
mov r1, 5

strex r2, r1, [r0]
cmp r2, 0
bne do_something_to_recover
```

`strex` tries to store `r1` in the memory that `r0` is pointing to, but checks the exclusive monitor first.

If the store is allowed, `r2` is set to 0, if it fails then `r2` is set to 1.

Then what should we do?

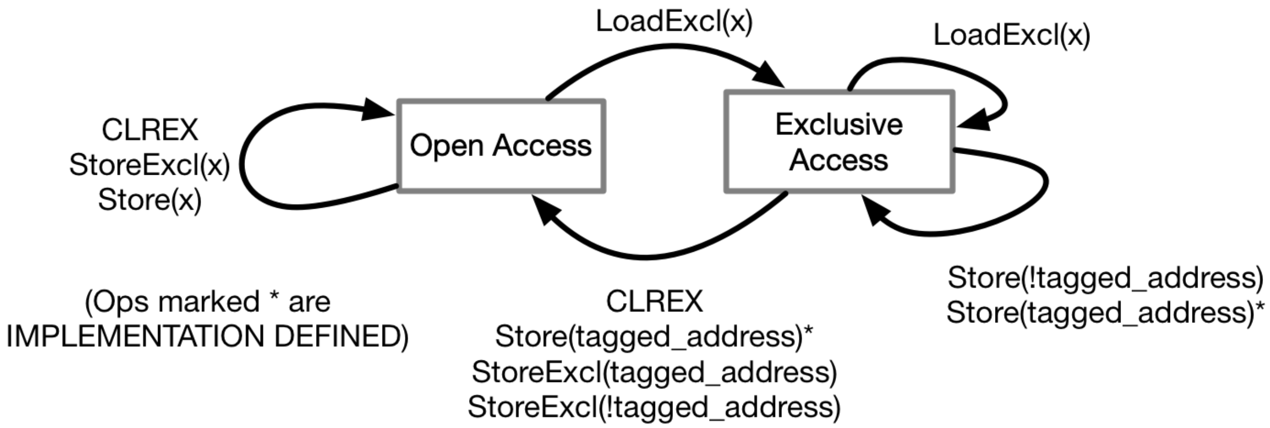
When can a `strex` actually fail?

Need to look in the ARMv7-M reference manual (Section A3.4 “Synchronisation and Semaphores”)

1. If address of `strex` is tagged exclusive in the local monitor; then store takes place (woo hoo!)
2. If address of `strex` is NOT tagged exclusive; then it is “implementation defined” whether the store takes place (????).

Local Exclusive Monitor

"Any `ldrex` operation updates the tagged address to the most significant bits of the address… used for the operation." (ARMv7-M reference manual)



Note that `clrex` **always** clears the monitor, and that interrupt handlers run `clrex`!

Let's experiment to see how this works.

Pt. 2: What do all these synth words mean?

Sounds?

How do we make interesting sounds with the disco boards?

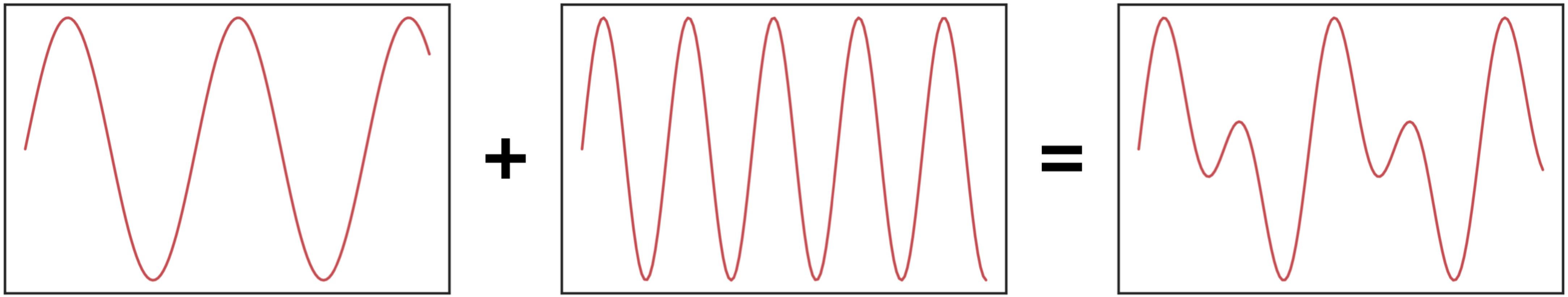
Digital Synthesis Approaches

- Additive synthesis
- Subtractive synthesis
- FM (frequency modulation) synthesis
- Envelopes and ADSR
- Drums: noise and non-pitched sounds
- Wavetable synthesis

What’s an Oscillator?

- A **module** (physical or code) that outputs a **waveform**.
- In synth lingo, sometimes a VCO (voltage controlled oscillator).

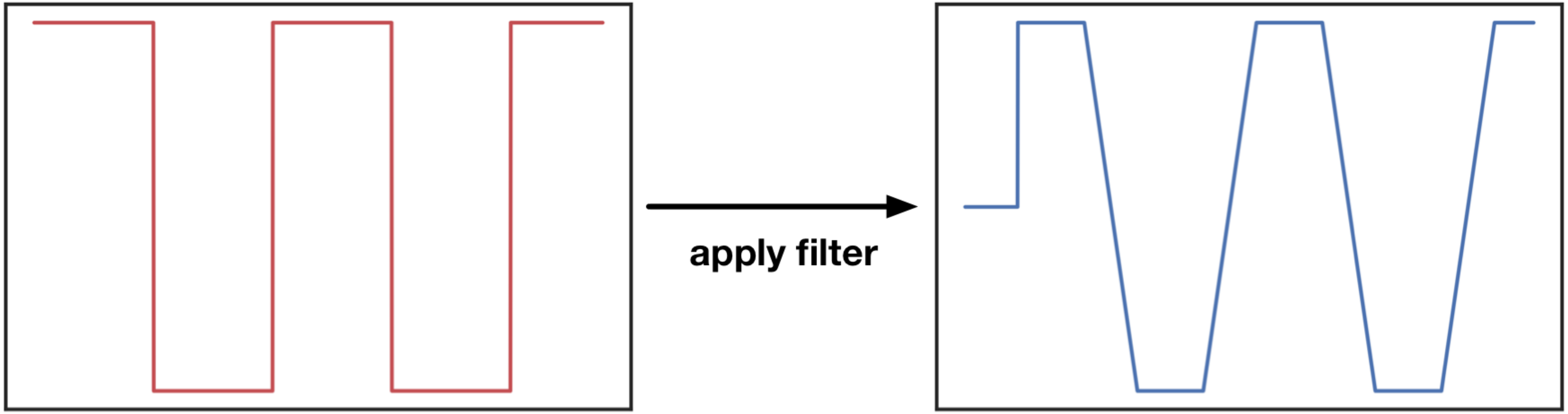
Additive Synthesis



$\sin(t)$ $\qquad$ $\sin(2t)$ $\qquad$ $\sin(t) + \sin(2t)$

- Take multiple oscillators and add them together!

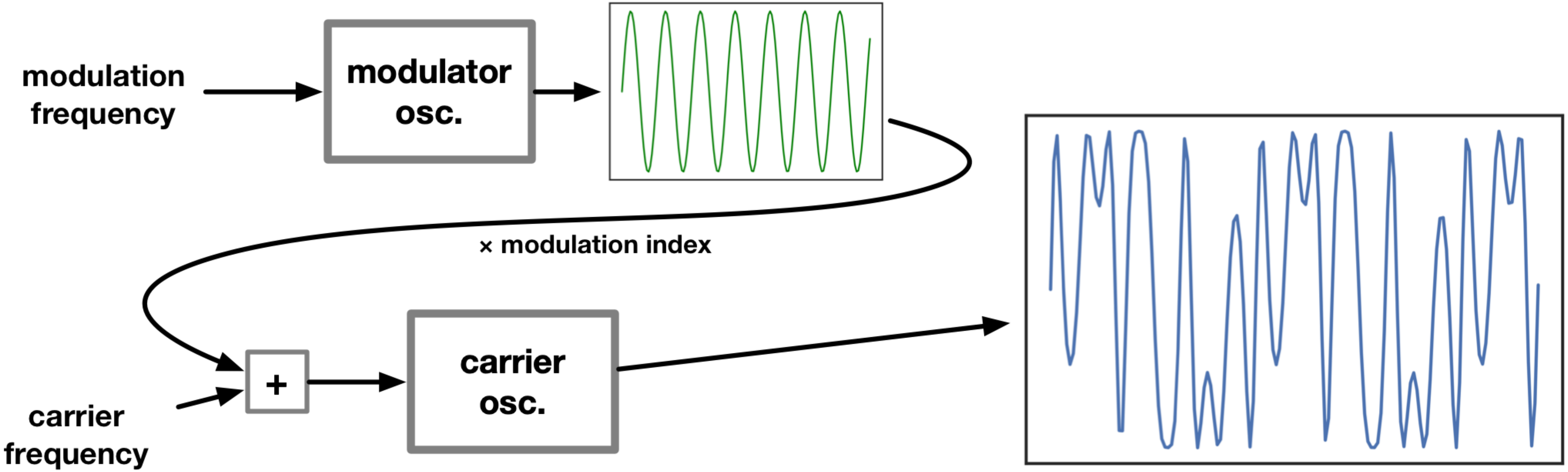
Subtractive Synthesis



n-sample moving average filter

- Use one oscillator and *take sound away*.

FM synthesis



$$\sin(ct + d\sin(mt))$$

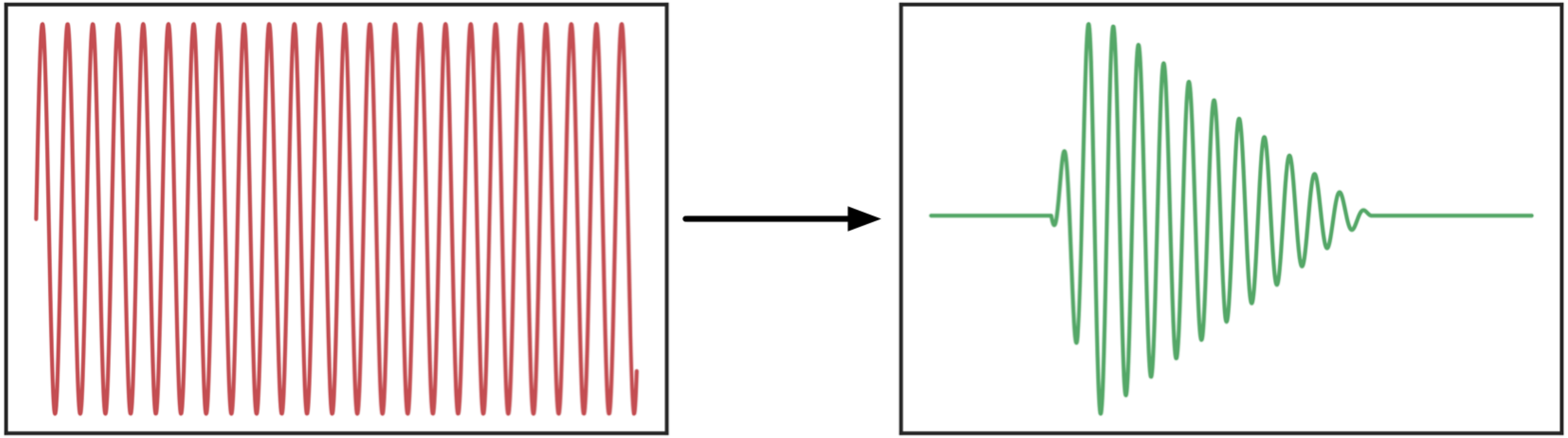
- “frequency modulation”

talk

So far we've made "sounds", but we want to make "notes".

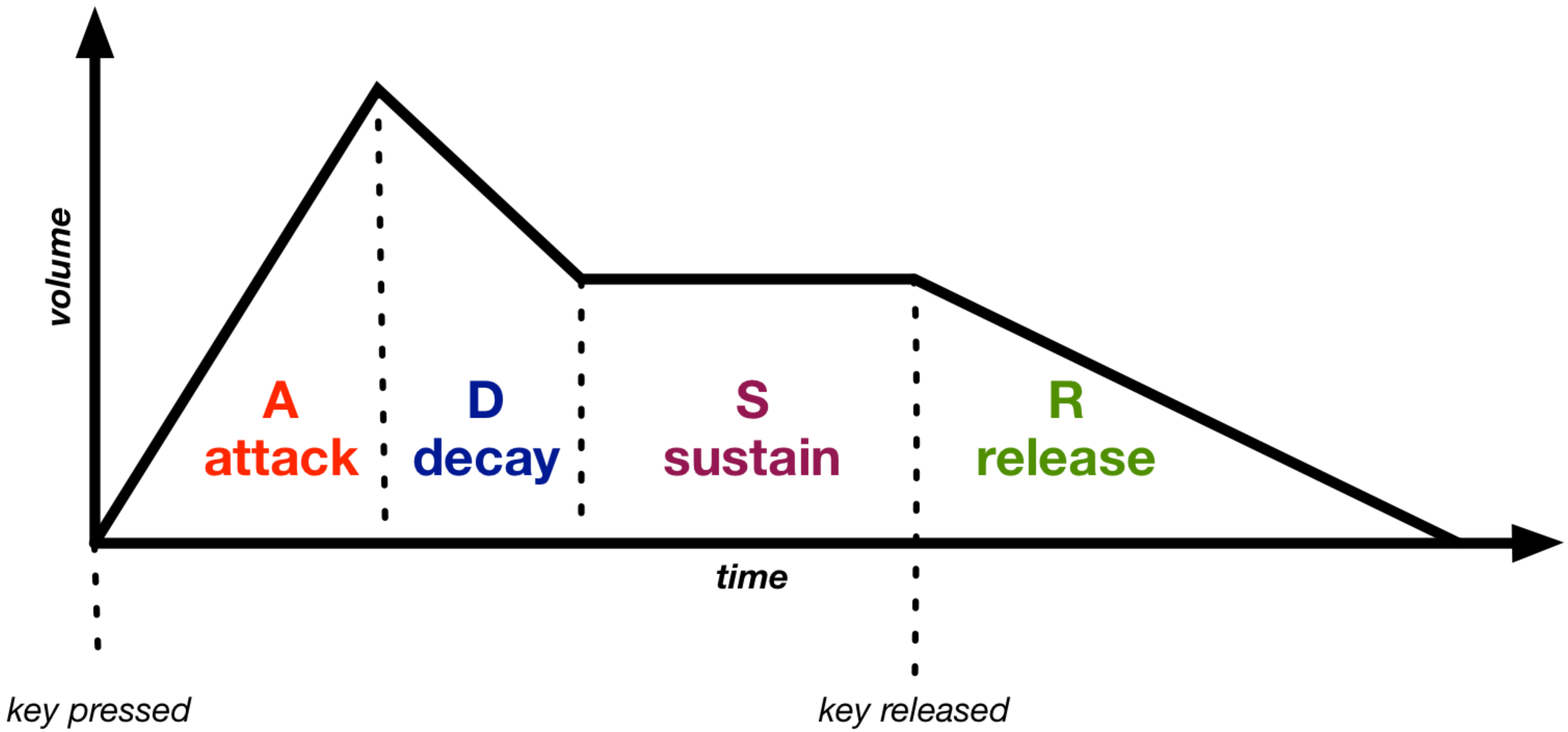
How can we do that?

Amplitude Envelope



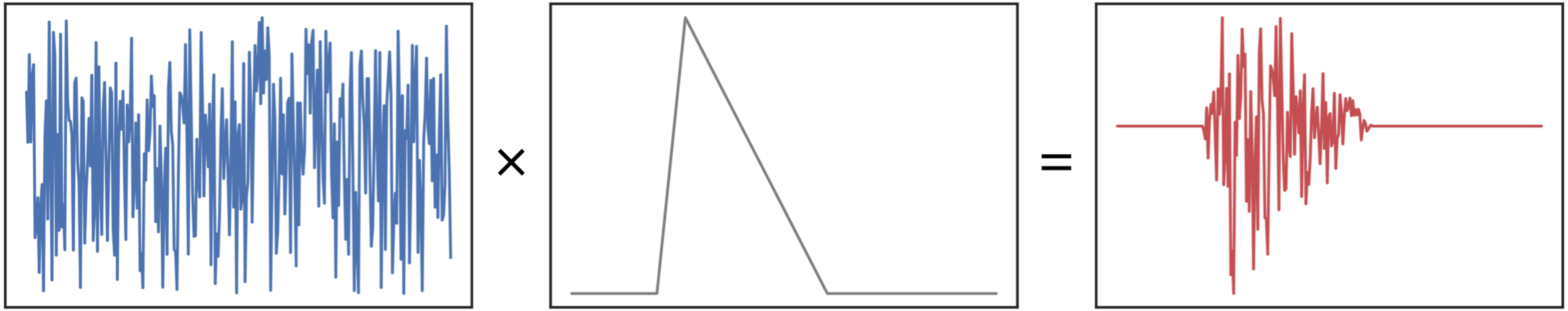
- **Amplitude** is the “volume” of our note.

ADSR Envelope



- The **adsr** shape is often used for pitched sounds.

Drums and Percussion



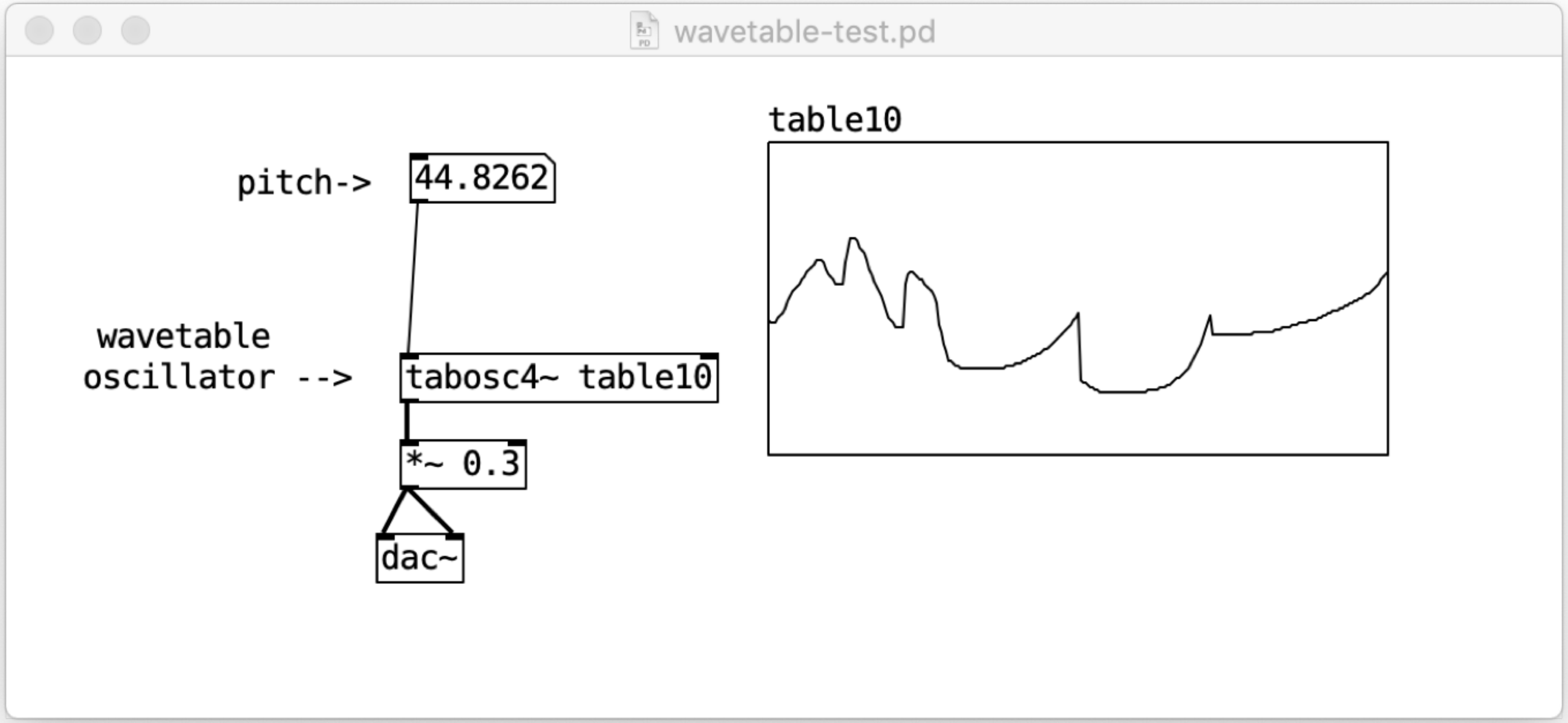
uniform noise

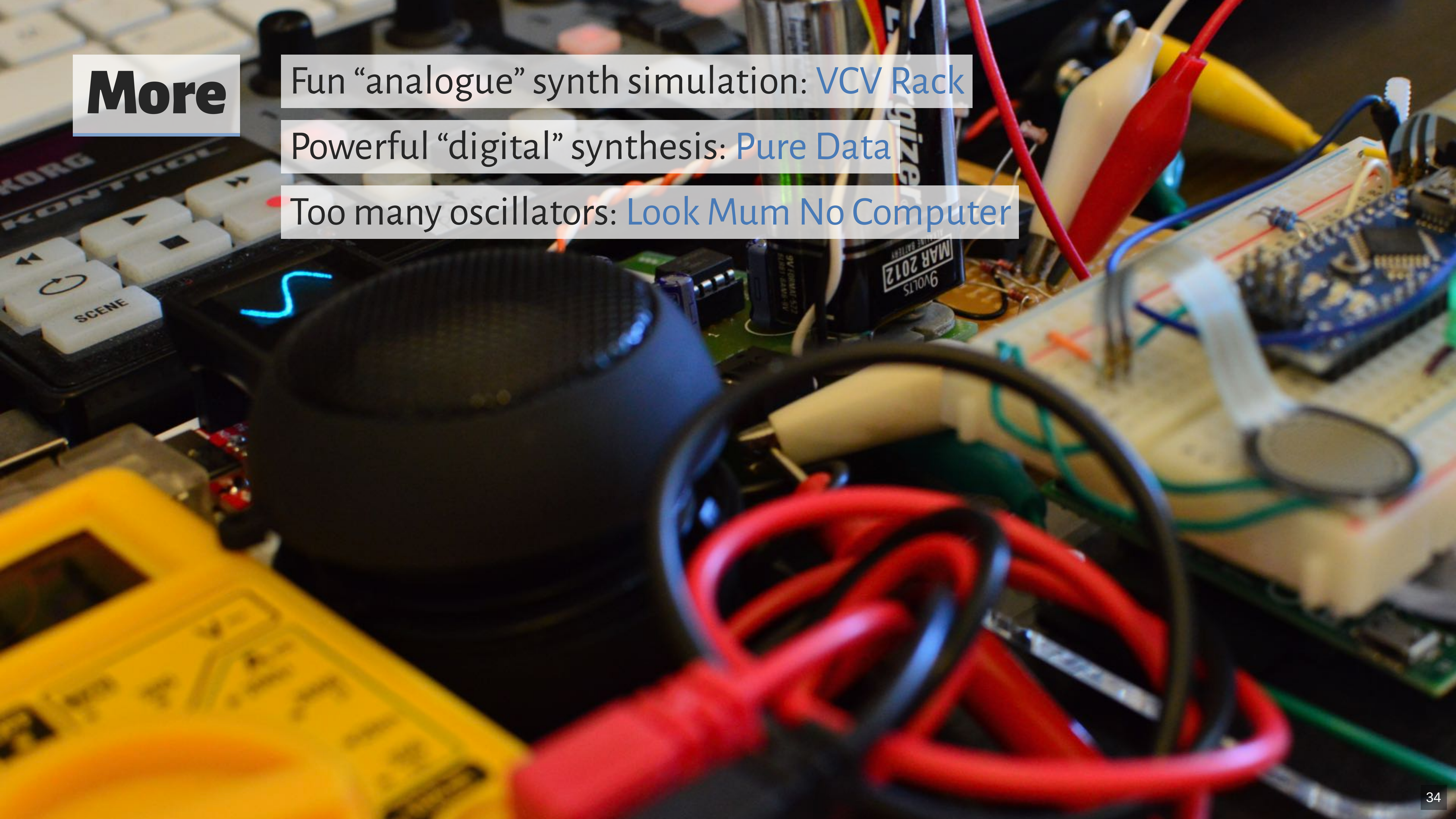
attack-release envelope

"Pssht!"

- Are drums all "non-pitched" sounds?

Wavetable Synthesis





More

Fun “analogue” synth simulation: VCV Rack

Powerful “digital” synthesis: Pure Data

Too many oscillators: Look Mum No Computer

questions?